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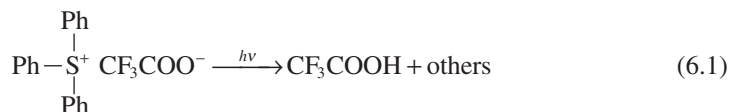
Chemically Amplified Resists: Exposure and Bake Chemistry

Unlike conventional resists, such as the diazonaphthoquinone (DNQ)/novolac systems discussed in the previous chapter, chemically amplified resists require two separate chemical reactions in order to change the solubility of the resists. First, exposure turns an aerial image into a latent image of exposure reaction products. Although very similar to conventional resists, the reaction products of exposure for a chemically amplified resist do not appreciably change the solubility of the resist. Instead, a second reaction during a post-exposure bake is catalyzed by the exposure reaction products. The result of the post-exposure bake reaction is a change in the solubility of the resist. This two-step process has some interesting characteristics and challenges.

In many respects, chemically amplified resists are a conceptual superset of conventional resists. With the exception of thermal decomposition, all of the concepts discussed in the previous chapter apply to chemically amplified resists as well. Familiarity with the concepts of that chapter is thus assumed in this chapter.

6.1 Exposure Reaction

Chemically amplified photoresists are composed of a polymer resin (possibly ‘blocked’ to inhibit dissolution), a sensitizer called a *photoacid generator* (PAG), and possibly a cross-linking agent, dye or other additive. As the name implies, the PAG forms an acid when exposed to deep-UV light. Ito and Willson at IBM first proposed the use of an aryl onium salt,¹ and triphenylsulfonium salts have been studied extensively as PAGs. The reaction of a very simple triphenylsulfonium salt PAG is:



The acid generated in this case (trifluoroacetic acid) is a derivative of acetic acid where the electron-drawing properties of the fluorines greatly increase the acidity of the molecule. In fact, very strong acids will be required for the chemical amplification step described in the next section. The PAG is mixed with the polymer resin at a concentration of typically 5–15% by weight for 248-nm resists, with 10% as a typical formulation (corresponding to a PAG concentration of about 0.2 mol/liter). For 193-nm resists, PAG loading is kept lower at 1–5 wt% in order to keep the optical absorbance of the resist within desired levels.

The kinetics of the exposure reaction are presumed to be standard first order (just as for the exposure of PAC in a conventional resist):

$$\frac{\partial G}{\partial t} = -CIG \quad (6.2)$$

where G is the concentration of PAG at exposure time t (the initial PAG concentration is G_0), I is the exposure intensity, and C is the exposure rate constant. For most chemically amplified resists, the bleachable absorption coefficient A is zero and the intensity in the resist stays constant during exposure. For this case, the rate equation can be solved for G :

$$G = G_0 e^{-CIt} \quad (6.3)$$

The acid concentration H is given by

$$H = G_0 - G = G_0(1 - e^{-CIt}) \quad (6.4)$$

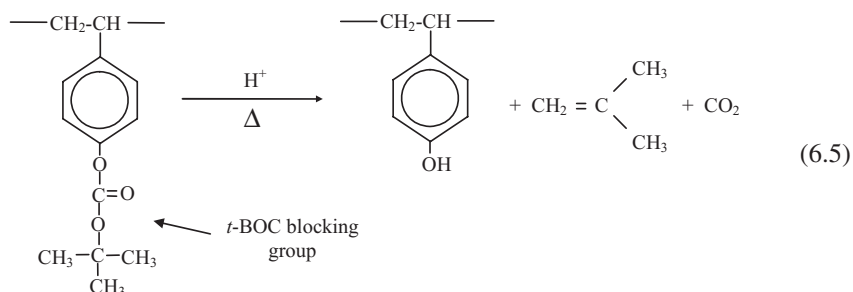
Some 248-nm PAGs react to exposure by a different mechanism. The polymer resin used by the resist (such as polyhydroxystyrene, discussed below) will absorb light, resulting in electrons promoted to higher energy states in the polymer. It is possible for some of this energy to be transferred from the polymer to the PAG, providing an indirect pathway for photon absorption to provide energy to the PAG. If this mechanism exists in a resist, the resulting kinetics will still be well described by the first-order kinetics discussed above.

6.2 Chemical Amplification

Exposure of the resist with an aerial image $I(x)$ results in an acid latent image $H(x)$. A post-exposure bake (PEB) is then used to thermally induce a chemical reaction. This may be the activation of a cross-linking agent for a negative resist or the deblocking of the polymer resin for a positive resist. The defining characteristic of a chemically amplified resist is that this reaction is *catalyzed* by the acid so that the acid is not consumed by the reaction and, to first order, H remains constant. As a result, each exposure event (creating one acid molecule) can cause a large number of dissolution-changing chemical events during PEB. The effects of exposure are said to be ‘amplified’ by the catalytic nature of this chemical reaction. The *catalytic chain length*, defined to be the average number of amplification chemical events caused by one acid molecule, is in the range of 10 to 100 for most chemically amplified resists, with 20 a typical catalytic chain length. Chemically amplified resists use exposure doses in the range of 20–50 mJ/cm², making them 5–10 times more sensitive than the DNQ resists used at i-line.

6.2.1 Amplification Reaction

Willson, Ito and Frechet first proposed the concept of deblocking a polymer to change its solubility.² A base polymer, such as polyhydroxystyrene (PHS), is used that is very soluble in an aqueous base developer. It is the acidic hydroxyl (—OH) groups that give the PHS its high solubility, so by ‘blocking’ these sites (by reacting the hydroxyl group with some longer-chain molecule) the solubility can be reduced. The IBM team employed a *t*-butoxycarbonyl group (*t*-BOC), resulting in a very slowly dissolving polymer. In the presence of a strong acid and heat, the *t*-BOC blocked polymer will undergo acidolysis to generate the soluble hydroxyl group, as shown below.



One drawback of this scheme is that the cleaved *t*-BOC is volatile and will evaporate, causing film shrinkage in the exposed areas. Larger molecular weight blocking groups are commonly used to reduce this film shrinkage to acceptable levels (below 10%). Also, the blocking group is such an effective inhibitor of dissolution that nearly every blocked site on the polymer must be deblocked to obtain significant dissolution. Thus, the photoresist can be made more ‘sensitive’ by only partially blocking the PHS (creating what is called a copolymer, since the polymer chain is now made up of two repeating units – blocked and unblocked PHS). Additionally, fully blocked polymers tend to have poor coating and adhesion properties. Typical photoresists use 10–30% of the hydroxyl groups blocked, with 20% as a typical value. Molecular weights for the PHS run in the range of 3000–6000 giving about 20–40 hydroxyl groups per polymer molecule, about 4 to 10 of which are initially blocked in typical formulations. As an example, the early chemically amplified resist Apex-E has a molecular weight of about 5000, giving about 35 hydroxyl sites per polymer, about 25% of which are initially blocked by *t*-BOC.

This deblocking reaction (also called the deprotection reaction) is often characterized as being either low activation or high activation, referring to the amount of heat that must be supplied to make the reaction go. For a low activation resist, deblocking can occur at room temperature, and so begins during the exposure process. For a high activation resist, a post-exposure bake in the 95–135 °C temperature range is required. While low activation resists have an advantage in not being sensitive to delay time effects (see section 6.2.3), high activation resists tend to be more readily controlled (produce more uniform resist feature sizes) and are thus the most commonly used. Typical high and low activation blocking groups are shown in Figure 6.1. Many resists today in fact include copolymers of both high and low activation blocking groups.

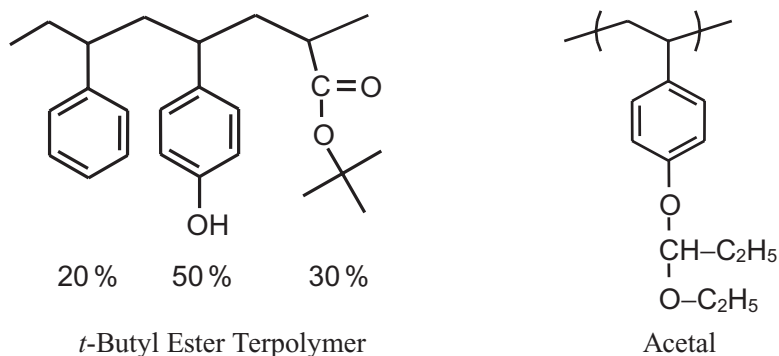


Figure 6.1 Examples of 248-nm blocking groups: the high activation *t*-butyl ester; and the low activation acetal

Using M as the concentration of some reactive site (such as the *t*-BOC blocking group), these sites are consumed (i.e. are reacted) according to kinetics of first order in H and first order in M :

$$\frac{\partial M}{\partial t_{\text{PEB}}} = -k_4 MH \quad (6.6)$$

where k_4 is the rate constant of the amplification reaction (cross-linking, deblocking, etc.) and t_{PEB} is the bake time. Assuming that H is constant (an assumption that will be eliminated later in this chapter), Equation (6.6) can be solved:

$$M = M_0 e^{-k_4 H t_{\text{PEB}}} \quad (6.7)$$

(Note: Although H^+ is not consumed by the reaction, the value of H is not locally constant. Diffusion during the PEB and acid loss mechanisms cause local changes in the acid concentration, thus requiring the use of a reaction–diffusion system of equations as discussed below. The approximation that H is constant is a useful one, however, that gives insight into the reaction as well as accurate results under some conditions.)

It is useful here to normalize the concentrations to some initial values. This results in a normalized acid concentration h and normalized unreacted sites m :

$$h = \frac{H}{G_0} \quad m = \frac{M}{M_0} \quad (6.8)$$

Equations (6.4) and (6.7) become

$$\begin{aligned} h &= 1 - e^{-Ct} \\ m &= e^{-K_{\text{amp}} t_{\text{PEB}} h} \end{aligned} \quad (6.9)$$

where $K_{\text{amp}} = G_0 k_4$. The result of the PEB is an amplified latent image $m(x)$, corresponding to an exposed latent image $h(x)$, resulting from the aerial image $I(x)$. The amount of amplification (the conversion of an exposed latent image of acid into an amplified latent image of blocked and deblocked polymer) is a function of PEB time and temperature. The time dependence in Equation (6.9) is obvious, and the temperature dependence comes through the amplification rate constant K_{amp} .

Table 6.1 Typical values for amplification reaction activation energies as a function of blocking group

Blocking Group (248-nm resists)	Activation Energy (kcal/mol)
<i>t</i> -BOC	30
<i>t</i> -Butyl Ester	22
Acetal	15

Like most rate constants, the temperature dependence of K_{amp} can be described by an Arrhenius relation:

$$K_{\text{amp}} = A_r e^{-E_a/RT} \quad (6.10)$$

where A_r is the Arrhenius coefficient for this reaction (1/s), E_a is the activation energy (cal/mole or J/mole), R is the universal gas constant (1.98717 cal/mole-K or 8.31431 J/mole-K), and T is the absolute temperature (K). Typical values for activation energies as a function of blocking group are given in Table 6.1. The resulting K_{amp} at the PEB temperature tends to be in the range of 0.02–0.1 s⁻¹.

Equations (6.9) reveal an interesting symmetry between exposure dose, given by It , and a kind of ‘thermal dose’. Consider the case of a small exposure dose ($It \ll 1/C$) so that the amount of acid generated is small. Equations (6.9) will become

$$\begin{aligned} h &\approx CIt \\ m &\approx e^{-K_{\text{amp}} t_{\text{PEB}} CIt} \end{aligned} \quad (6.11)$$

If we define the effective thermal dose as $K_{\text{amp}} t_{\text{PEB}}/C$, then there exists a reciprocity between exposure dose and thermal dose. Cutting exposure dose in half and doubling the thermal dose will result in the same chemical change at the end of the amplification reaction. As we shall see in the following section, however, the true picture is complicated by the effects of diffusion.

6.2.2 Diffusion

The above analysis of the kinetics of the amplification reaction assumed a locally constant concentration of acid H . Although this could be exactly true in some circumstances, it is typically only an approximation, and is often a poor approximation. In reality, the acid diffuses during the bake. The standard diffusion equation takes the form

$$\frac{\partial H}{\partial t_{\text{PEB}}} = \nabla(D_H \nabla H) \quad (6.12)$$

where D_H is the diffusivity of acid in the photoresist. Solving this equation requires a number of things: two boundary conditions for each dimension, one initial condition and knowledge of the diffusivity as a function of position and time.

The solution of Equation (6.12) can now be performed if the diffusivity of the acid in the photoresist is known. Unfortunately, this solution is complicated by two very important factors: the diffusivity is a strong function of temperature and, most probably, the

extent of amplification. Since the temperature is changing with time during the bake, the diffusivity will be time dependent. The concentration dependence of diffusivity results from an increase in free volume for typical positive resists: as the amplification reaction proceeds, the polymer blocking group evaporates resulting in a decrease in film thickness but also an increase in free volume (and probably a change in the glass transition temperature as well). Since the acid concentration is time and position dependent, the diffusivity in Equation (6.12) must be determined as a part of the solution of Equation (6.12) by an iterative method. The resulting simultaneous solution of Equations (6.7) and (6.12) is called a *reaction–diffusion* system.

The temperature dependence of the diffusivity can be expressed in a standard Arrhenius form:

$$D_0(T) = A_r e^{-E_a/RT} \quad (6.13)$$

where D_0 is a general diffusivity. A full treatment of the amplification reaction would include a thermal model of the hot plate in order to determine the actual time–temperature history of the wafer (see Chapter 5). To simplify the problem, an ideal temperature distribution can be assumed – the temperature of the resist is zero (low enough for no diffusion or reaction) until the start of the bake, at which time it immediately rises to the final bake temperature, stays constant for the duration of the bake, then instantly falls back to zero.

The concentration dependence of the diffusivity is less obvious. Several authors have proposed and verified the use of different models for the concentration dependence of diffusion within a polymer. Of course, the simplest form (besides a constant diffusivity) would be a linear model. Letting D_0 be the diffusivity of acid in completely unreacted resist and D_f the diffusivity of acid in resist that has been completely reacted,

$$D_H = D_0 + (D_f - D_0)(1 - m) \quad (6.14)$$

Here, diffusivity is expressed as a function of the extent of the amplification reaction $1 - m$. Another common form is the Fujita–Doolittle equation (see Chapter 5), which can be predicted theoretically using free volume arguments. A form of that equation that is convenient for calculations is:

$$D_H = D_0 \exp\left(\frac{D_2(1 - m)}{1 + D_3(1 - m)}\right) \quad (6.15)$$

where $D_2 = (1 + D_3)\ln(D_f/D_0)$ and D_3 is an experimentally determined constant. Other concentration relationships are also possible.

While general solutions to these reaction–diffusion equations require numerical techniques, there are some special cases where analytical solutions are possible. Consider a constant acid diffusivity and an ideal temperature profile (that is, ignore the ramp up to the bake temperature). Further, ignore any sources of acid loss (to be discussed in the next section). Then, the solution to the reaction rate equation given in Equation (6.9) can be used if the relative acid concentration is replaced by an effective acid concentration, given by

$$h_{\text{eff}}(x) = \frac{1}{t_{\text{PEB}}} \int_0^{t_{\text{PEB}}} h(x, t = 0) \otimes \text{DPSF} \, dt \quad (6.16)$$

where $h(x, t = 0)$ is the acid concentration at the beginning of the bake and t_{PEB} is the PEB bake time. Diffusion is given by the convolution of the acid profile with the diffusion point spread function (*DPSF*, defined in Chapter 5). The effective acid concentration at a particular x -position is the average concentration at that position over time. This equation can be further arranged to define a reaction–diffusion point spread function, *RDPSF*:

$$h_{\text{eff}}(x) = h(x, 0) \otimes \text{RDPSF}$$

where

$$\text{RDPSF} = \frac{1}{t_{\text{PEB}}} \int_0^{t_{\text{PEB}}} \text{DPSF} \, dt \quad (6.17)$$

The Gaussian diffusion kernel is affected by time integration through the diffusion length, $\sigma_D = \sqrt{2Dt}$, where D is the acid diffusivity in resist. Thus, using the 1D case as an example,

$$\text{RDPSF} = \frac{1}{t_{\text{PEB}} \sqrt{4\pi D}} \int_0^{t_{\text{PEB}}} \frac{e^{-x^2/4Dt}}{\sqrt{t}} \, dt \quad (6.18)$$

The integral is solvable, resulting in an interesting final solution.

$$\text{RDPSF}(x) = 2 \frac{e^{-x^2/2\sigma_D^2}}{\sqrt{2\pi}\sigma_D} - \frac{|x|}{\sigma_D^2} \text{erfc}\left(\frac{|x|}{\sqrt{2}\sigma_D}\right) \quad (6.19)$$

The first term on the right hand side of Equation (6.19) is nothing more than twice the *DPSF*, and thus accounts for pure diffusion. The second term, the complimentary error function times x , is a reaction term that is subtracted and thus reduces the impact of pure diffusion. Each term, as well as the final *RDPSF*, is plotted in Figure 6.2.

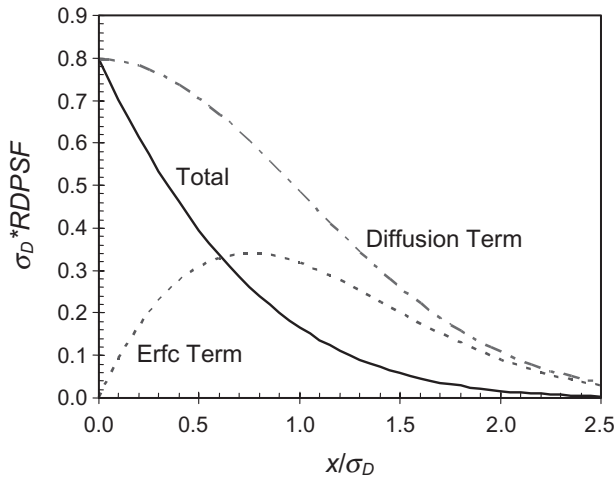


Figure 6.2 The 1D reaction–diffusion point spread function (*RDPSF*) and its component terms

Equation (6.19) gives the *RDPSF* in one dimension. Extension to two or three dimensions is straightforward:

$$\begin{aligned} \text{2D: } RDPSF(r) &= -\frac{1}{\pi\sigma_D^2} Ei\left(-\frac{r^2}{2\sigma_D^2}\right), \quad r = \sqrt{x^2 + y^2} \\ \text{3D: } RDPSF(r) &= \frac{1}{2\pi\sigma_D^3} \left[\frac{\sigma_D}{|r|} \operatorname{erfc}\left(\frac{|r|}{\sqrt{2}\sigma_D}\right) \right], \quad r = \sqrt{x^2 + y^2 + z^2} \end{aligned} \quad (6.20)$$

where *Ei* is the exponential integral. Once the effective acid concentration is known, the amount of deblocking can be calculated as before with a simple modification to Equation (6.9):

$$m = e^{-K_{\text{amp}} t_{\text{PEB}} h_{\text{eff}}} \quad (6.21)$$

6.2.3 Acid Loss

Through a variety of mechanisms, acid formed by exposure of the resist film can be lost and thus not contribute to the catalyzed reaction to change the resist solubility. There are two basic types of acid loss – loss that occurs between exposure and post-exposure bake, and loss that occurs during the post-exposure bake. The first type of loss leads to delay time effects – the resulting lithographic patterns are affected by the delay time between exposure and post-exposure bake. Delay time effects can be very severe and, of course, are very detrimental to the use of such a resist in a manufacturing environment. Acid loss during PEB may be detrimental (if, for example, it is nonuniform or affects only the top or the bottom of the resist profile), or it may serve the beneficial function of stopping the reaction and limiting the sensitivity of the final feature size to bake time or temperature.

The typical mechanism for delay time acid loss is the diffusion of atmospheric base contaminants into the top surface of the resist. The result is a neutralization of the acid near the top of the resist and a corresponding reduced amplification. For a negative resist, the top portion of a line is not insolubilized and resist is lost from the top of the line. For a positive resist, the effects are more devastating. Sufficient base contamination can make the top of the resist insoluble, blocking dissolution into the bulk of the resist (Figure 6.3)

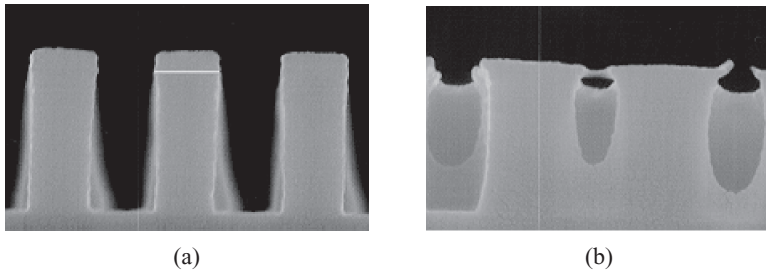


Figure 6.3 Atmospheric base contamination leads to T-top formation. Shown are line/space features printed in APEX-E for: (a) 0.275- μm features with no delay; and (b) 0.325- μm features with 10-minute delay between exposure and post-exposure bake (courtesy of SEMATECH)

and creating a characteristic ‘T-top’-shaped resist profile. In extreme cases, no patterns are observed after development.

The effects of acid loss due to atmospheric base contaminants can be accounted for in a straightforward manner. The base diffuses from the top surface of the resist into the bulk. Assuming that the concentration of base contaminant in contact with the top of the resist remains constant, the diffusion equation can be solved for the concentration of base, B , as a function of depth into the resist film:

$$B = B_0 \operatorname{erfc}\left(\frac{z}{\sqrt{2}\sigma_B}\right) \quad (6.22)$$

where B_0 is the base concentration at the top of the resist film, z is the depth into the resist ($z = 0$ at the top of the film) and σ_B is the diffusion length of the base in resist. B_0 will be directly proportional to the concentration of base in the atmosphere by Henry’s law (see Problem 6.8). The standard assumption of constant diffusivity has been made here so that the diffusion length goes as the square root of the delay time and as the square root of the diffusivity. Note that this equation assumes that during the post-exposure delay period, the base diffusion length is much greater than the photogenerated acid diffusion length. Since the acid generated by exposure for most resist systems of interest is fairly strong, it is a good approximation to assume that all of the base contaminant will react with acid if there is sufficient acid present.

Atmospheric base contamination can be devastating to chemically amplified resist performance because of the very nature of chemical amplification. Since one photo-generated acid causes dozens of dissolution changing events during PEB, loss of even a small amount of acid before the PEB will have an amplified effect. Several different countermeasures are usually taken to reduce the impact of atmospheric base. As Equation (6.22) shows, base contamination in the resist can be reduced by reducing the atmospheric base (B_0) or by reducing the base diffusion length. In turn, diffusion length can be reduced by reducing the base diffusivity or by reducing the delay time. In practice, all three of these techniques are commonly used. Photolithography equipment is placed as far as possible from sources of base contaminants (resist strippers and ammonia being common examples). Imaging and resist processing tools are then enclosed and the air is filtered with activated charcoal. The delay time between exposure and post-exposure bake is kept to a minimum by directly linking the resist processing track to the step and scan exposure tool. Each wafer is sent to be baked immediately after its exposure is complete. Finally, the diffusivity of base in the resist is reduced by ‘annealing’ the resist. If the post-apply bake temperature is set to be above the glass transition temperature of the resist polymer, solvent and free volume will be annealed out of the resist (see Chapter 5). It is this free volume that enables the diffusion of base into the resist, so elimination of free volume greatly reduces the contaminant diffusivity.

Another possible acid loss mechanism is base contamination from the substrate, as has been observed on TiN and other nitrogen-containing substrates. These substrates are thought to trap acid that diffuses down to the substrate, resulting in resist footing for a positive resist. Plasma and chemical treatments of nitrogen containing substrates can pacify trapping sites, but the common use of an organic bottom antireflection coating (BARC) generally eliminates the possibility of substrate interactions. Acid can also be

lost at the top surface of the resist due to evaporation. The amount of evaporation is a function of the size of the acid and the degree of its interaction with the resist polymer. A small acid (such as the trifluoroacetic acid discussed above) may have very significant evaporation. Thus, most resists today employ fairly large acid molecules to avoid significant acid evaporation.

6.2.4 Base Quencher

Another mechanism for acid loss is intentional rather than accidental. Most modern formulations of chemically amplified resists include the addition of a base quencher (compounds such as tetraoctylammonium hydroxide, for example³). Loaded at concentrations of 5–20% of the initial PAG loading, this base quencher (Q) is designed to neutralize any photogenerated acid that comes in contact with it.



Acid–base neutralization reactions tend to be equilibrium reactions, but in this case the equilibrium constant K_{eq} is large due to the strength of the acid, meaning the reaction favors heavily the formation of the salt HQ . Thus, it is a good approximation to ignore the reversible nature of the reaction and replace the equilibrium constant with a standard forward reaction rate constant, k_Q .

For low exposure doses, the small amount of photoacid generated will be neutralized by the base quencher and amplification will not take place (Figure 6.4). Only when the exposure rises above a certain threshold will the amount of acid be sufficient to completely neutralize all of the base quencher and have leftover acid that can cause deblocking during PEB. The main purpose of the base quencher is to neutralize the low levels of acid that might diffuse into the nominally unexposed regions of the wafer, thus making the final resist linewidth less sensitive to acid diffusion. In addition, base quencher will reduce the sensitivity of the resist to airborne base contaminants.

The simple description of base quenching behavior above is made more complicated by the fact that the quencher will, in general, diffuse during the post-exposure bake. Base quencher diffusion can take on any of the concentration and temperature dependencies described above for acid diffusion. The difference in diffusivity between the acid and the base becomes an important descriptor of lithographic behavior for these types of

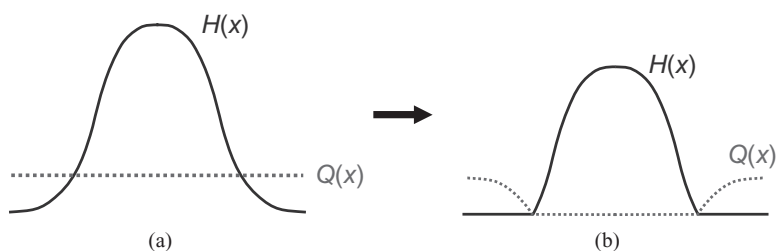


Figure 6.4 The effect of quencher (Q) on the acid latent image $H(x)$: (a) after exposure but before the quenching reaction; and (b) after the quenching reaction (assuming $K_{quench} \gg K_{amp}$)

resists. Rate equations for the acid and base, for the case of constant diffusivity, are given by

$$\begin{aligned}\frac{dh}{dt} &= -K_{\text{quench}}hq + D_H\nabla^2h \\ \frac{dq}{dt} &= -K_{\text{quench}}hq + D_Q\nabla^2q\end{aligned}\quad (6.24)$$

where $K_{\text{quench}} (= k_Q G_0)$ is the normalized quenching reaction rate constant and $q (= Q/G_0)$ is the normalized base quencher concentration. The impact of diffusion can be understood qualitatively by examining Figure 6.4b. If acid diffuses, but base does not, the zero points (the points at which the acid and base completely neutralize each other) will move toward the outside. If the base diffuses but the acid does not, the opposite will occur and the zero points will move in toward the center of the figure. There will be some combination of acid and base diffusion (diffusion lengths about equal) that will leave the zero point stationary. Note that diffusion is driven by the *change* in the gradient of the diffusing species (a second derivative), so that it is a combination of the diffusivities of the acid and the base and the shape of the latent image that determines which direction the zero point will move.

If diffusion is ignored and $K_{\text{quench}} \gg K_{\text{amp}}$, the impact of a base quencher is to first order a dose offset – one must generate enough acid to overcome the quencher before amplification can occur. Thus, the simple (nondiffusion) kinetic result of Equation (6.9) becomes

$$\begin{aligned}m &= e^{-K_{\text{amp}} t_{\text{PEB}} (h - q_0)} & \text{when } h > q_0 \\ m &= 1 & \text{when } h < q_0\end{aligned}\quad (6.25)$$

where the ‘0’ subscript refers to the initial condition (at the beginning of the PEB). Note that in the formulation of the above equation, h represents the relative acid concentration generated upon exposure, but before any quenching takes place.

The impact of base quencher is most readily seen by comparing latent images of resist formulations with and without quencher. When both resist formulations are exposed at the dose-to-size, with all other processing parameters the same, the latent image slope improves dramatically near the resist feature edge when quencher is added (Figure 6.5).

6.2.5 Reaction–Diffusion Systems

The combination of a reacting system and a diffusing system is called a reaction–diffusion system. The solution of such a system is the simultaneous solution of Equations (6.6) and (6.12) using Equation (6.4) as an initial condition and possibly Equation (6.14) or (6.15) to describe the reaction-dependent diffusivity. Of course, any or all of the acid loss mechanisms can also be included.

The coupling of reaction and diffusion can be made more explicit. The Byers–Petersen model for chemically amplified reactions⁴ is a superset of the simpler chemically amplified models discussed above. In this model, the amplification reaction is thought to occur by two sequential steps: first, acid diffuses to the reaction site; then, acid reacts at that site (an idea first described by the Polish physicist Marian von Smoluchowski⁵ in 1917). If the diffusion step is very fast, the overall rate is controlled by the rate of the reaction

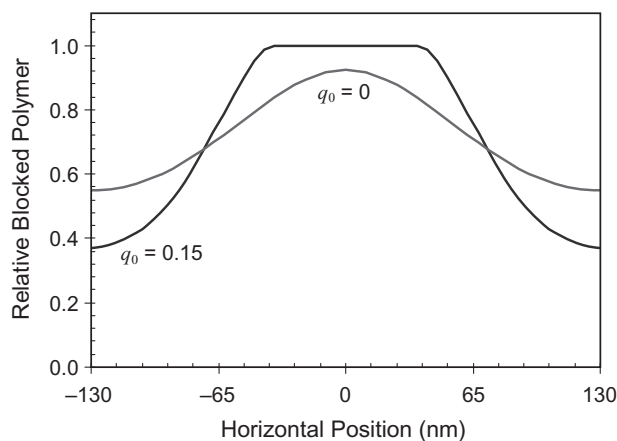


Figure 6.5 The effect of quencher on the shape of the latent image. Both resists have identical processing, except that the dose for each is adjusted to be the dose-to-size (130-nm lines and spaces)

itself (the amplification is said to be reaction-controlled). If, however, the diffusion is very slow, the overall rate of amplification will be controlled by the rate of diffusion of acid to the site (a diffusion-controlled amplification rate). If K_{react} is the rate constant for the reaction step, and $K_{\text{diff}}D_H$ is the rate constant (equal to the Smoluchowski trap rate normalized in the same way as K_{amp}) for the diffusion step, the overall amplification reaction rate constant will have the form

$$K_{\text{amp}} = \frac{K_{\text{react}}K_{\text{diff}}D_H}{K_{\text{react}} + K_{\text{diff}}D_H} \quad (6.26)$$

where D_H is the diffusivity of the acid.

Note that if K_{diff} is very large, then even for moderate values of acid diffusivity the kinetics will be reaction-controlled and Equation (6.26) will become

$$K_{\text{amp}} = K_{\text{react}} \quad (6.27)$$

and the Byers–Petersen model will revert to the original chemically amplified resist model given above. In general, the activation energies for K_{react} and for D_H will be different and it is possible for the overall reaction to switch from diffusion-controlled to reaction-controlled as a function of PEB temperature. For example, if the activation energy for the diffusion portion of the reaction is much smaller than the reaction activation energy, the overall reaction will be reaction-controlled at lower temperatures and diffusion-controlled at higher temperatures (Figure 6.6). K_{diff} represents the likelihood that the acid will be captured by a blocked site so that the reaction can occur. In the diffusion-controlled region, K_{react} is very high and

$$K_{\text{amp}} = K_{\text{diff}}D_H = 4\pi D_H a G_0 N_A \quad (6.28)$$

where a is the capture distance for the deblocking reaction (once the acid approaches the reaction site closer than the capture distance, a deblocking reaction is possible).

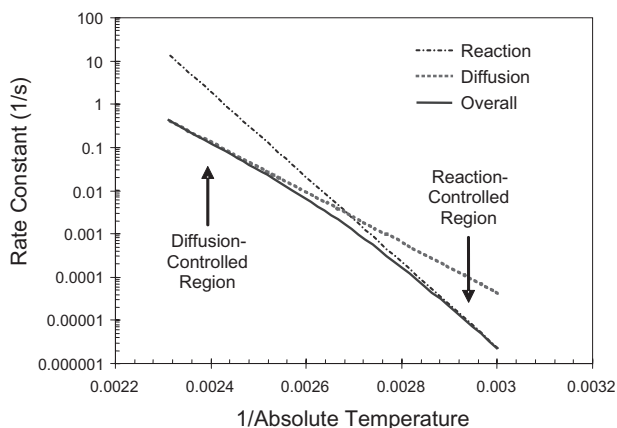


Figure 6.6 An Arrhenius plot (log of the rate constant versus the inverse of absolute temperature) for K_{amp} when the diffusion and amplification rate constants have different activation energies (diffusion activation energy = 26.5 kcal/mol; reaction activation energy = 45 kcal/mol)

As was discussed at the end of section 6.2.1, there is a trade-off in chemically amplified resists between exposure dose and thermal dose. Ignoring diffusion and acid loss, there is an approximate reciprocity between the two forms of dose. Generally, thermal dose is increased by an increase in the bake temperature (for production throughput reasons, PEB time is generally varied only over a small window). However, as the above discussion has shown, diffusion also varies exponentially with PEB temperature. Since there is an upper limit to the desired diffusion length of acid (and of base quencher), there is a practical upper limit to the temperature that can be used. And because of the exponential nature of the thermal dose with temperature, there is a practical minimum temperature that still provides for a low-enough dose to give good exposure tool throughput. Typically, the useful range of PEB temperatures for chemically amplified resists is at most 20 °C.

6.3 Measuring Chemically Amplified Resist Parameters

Since most chemically amplified resists do not bleach, the traditional optical transmittance experiment for measuring the *ABC* parameters gives no information about *C*. Other approaches have been devised to measure *C* that take advantage of the strength of the acid being generated in a chemically amplified resist. Direct measurement of the acid concentration is possible using titration, or indirectly using fluorescence. Alternatively, resist formulations with varying amounts of added base can be used to measure acid generation by determining the dose required to overcome the added base.

Deblocking during the chemical amplification reaction leads to the generation of volatile components that evaporate out of the resist during PEB. This slightly unpleasant side effect of amplification can be used as a way to measure the kinetics of the amplification reaction.⁶ Evaporation of volatile components at the relatively high PEB temperatures

used allows the film to relax and shrink. For a large open-frame exposure, the thickness loss after PEB is directly proportional to the average amount of deblocking that takes place through the thickness of the film (a good BARC should be used to avoid standing waves). If the maximum amount of film shrinkage, corresponding to 100% deblocking of the polymer, can be measured, film shrinkage after PEB can be directly related to the fractional amount of blocked groups left. By measuring film shrinkage versus dose, both the quencher concentration and the amplification rate constant can be obtained. By repeating this measurement at different PEB temperatures, the activation energy of the amplification can also be measured.

Figure 6.7 shows an example of the PEB kinetics measured using the PEB thickness loss measurement technique. Initially, no thickness loss is observed because the doses are not large enough to overcome the base quencher present in the resist. Past the dose required to neutralize all of the base quencher, the blocked fraction decreases as an exponential function, with the rate of decrease dependent on the amplification rate constant (and thus on the temperature). This figure also illustrates one of the major difficulties of this technique. Because the PEB time is fixed, there is no guarantee that even at the highest temperatures and highest exposure doses the resist will approach 100% deblocking. It is quite difficult to ensure that the measurement of the 100% deblocking thickness loss used to calibrate this technique does in fact correspond to 100% deblocking.

A more direct method of measuring chemical kinetics during PEB involves the use of Fourier transform infrared (FTIR) spectroscopy. Absorbance peaks as a function of wavelength indicate specific bonds in the various chemical species in the resist. As bonds are broken and formed during amplification, peaks of the FTIR spectrum appear and disappear, providing a quantifiable measure of the extent of reaction. FTIR equipment mounted directly above the PEB hot plate can measure these spectra in real time during the bake. However, since silicon is mostly transparent to IR light, a backside wafer coating of metal is required to provide a reflected signal to detect,⁷ or a hole in the hot plate must be used for measurement in transmission mode.⁸

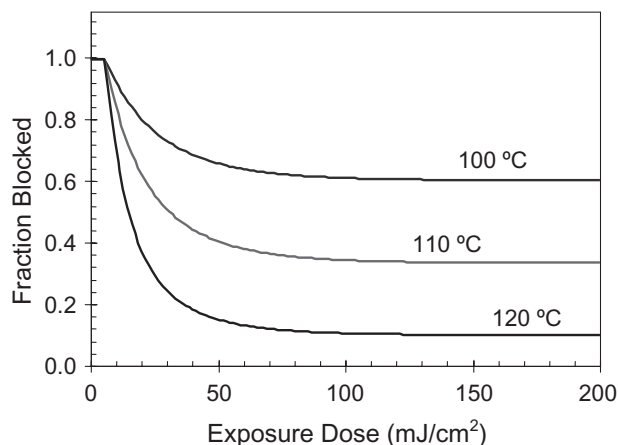


Figure 6.7 Example of PEB amplification kinetics measured by using thickness loss after PEB to estimate the fraction of polymer that remains blocked

The most difficult parameters to measure for chemically amplified resists are the acid and base diffusivities (the difficulties increase if these diffusivities vary with extent of amplification). In general, these parameters are empirically determined by matching predictions to experimental linewidth data.

6.4 Stochastic Modeling of Resist Chemistry

The theoretical descriptions of lithography given so far in this book make an extremely fundamental and mostly unstated assumption about the physical world being described: the so-called *continuum approximation*. Even though light energy is quantized into photons and chemical concentrations are quantized into spatially distributed molecules, the descriptions of aerial images and latent images ignore the discrete nature of these fundamental units and use instead continuous mathematical functions. For example, the very idea of chemical concentration assumes that the volume one is interested in is large enough to contain many, many molecules so that an average number of molecules per unit volume can be used. While we can mathematically discuss the idea of the concentration of some chemical species at a point in space, in reality this concentration must be an average extended over a large enough region. While in most cases the volumes of interest are large enough not to worry about this distinction, when trying to understand lithography down to the nanometer level the continuum approximation begins to break down.

When describing lithographic behavior at the nanometer level, an alternate approach, and in a very real sense a more fundamental approach, is to build the quantization of light as photons and matter as atoms and molecules directly into the models used. Such an approach is called *stochastic modeling*, and involves the use of random variables and probability density functions to describe the statistical fluctuations that are expected. Of course, such a probabilistic description will not make deterministic predictions – instead, quantities of interest will be described by their probability distributions, which in turn are characterized by their moments, such as the mean and variance.

6.4.1 Photon Shot Noise

Consider a light source that randomly emits photons at an average rate of L photons per unit time into some area A . Assume further that each emission event is independent. Over some small time interval dt (smaller than $1/L$ and small enough so that it is essentially impossible for two photons to be emitted during that interval), either a photon is emitted or it is not (a binary proposition). The probability that a photon will be emitted during this interval will be Ldt . Consider now some long time T ($\gg dt$). What can we expect for the number of photons emitted during the period T ? This basic problem is called a Bernoulli trial and the resulting probability distribution is the well-known *binomial distribution*. If $N = T/dt$, the number of time intervals in the total time, then the probability that exactly n photons will be emitted in this time period, $P(n)$, will be

$$P(n) = \binom{N}{n} (Ldt)^n (1 - Ldt)^{N-n} = \frac{N!}{(N-n)!n!} (Ldt)^n (1 - Ldt)^{N-n} \quad (6.29)$$

The binomial distribution is extremely cumbersome to work with as N gets large. If, however, $NLdt = TL$ remains finite as N goes to infinity, the binomial distribution converges to another, more manageable equation called the Poisson distribution:

$$\lim_{N \rightarrow \infty} \binom{N}{n} (Ldt)^n (1 - Ldt)^{N-n} = \frac{(TL)^n}{n!} e^{-TL} \quad (6.30)$$

Since there is no limit to how small dt can be made, letting dt go to zero will by default make N go to infinity for any nonzero time interval T and nonzero photon emission rate L . In general, the Poisson distribution is a good approximation to the binomial distribution whenever $TL > 1$ (that is, you expect to see more than one photon in the allotted time).

The Poisson distribution can be used to derive the statistical properties of photon emission. The expectation value of n [that is, the mean number of photons that will be emitted in a time interval T , denoted by the notation $E(n)$ or $\langle n \rangle$] is TL (a very reasonable result since L was defined as the average rate of photon emission):

$$\begin{aligned} \langle n \rangle \equiv E(n) &= \sum_{n=0}^{\infty} n P(n) \\ &= \sum_{n=0}^{\infty} n \frac{(TL)^n}{n!} e^{-TL} = e^{-TL} \sum_{n=1}^{\infty} \frac{(TL)^n}{(n-1)!} = TL e^{-TL} \sum_{m=0}^{\infty} \frac{(TL)^m}{m!} = TL \end{aligned} \quad (6.31)$$

The variance (the standard deviation squared) is also TL .

$$\sigma_n^2 \equiv E[(n - \langle n \rangle)^2] = E(n^2) - \langle n \rangle^2 = TL \quad (6.32)$$

To use these statistical properties, we must convert from number of photons to a more useful measure, intensity. If n photons cross an area A over a time interval T , the intensity of light will be

$$I = \frac{n}{TA} \left(\frac{hc}{\lambda} \right) \quad (6.33)$$

The mean value of this intensity will be

$$\langle I \rangle = \frac{\langle n \rangle}{TA} \left(\frac{hc}{\lambda} \right) = \frac{L}{A} \left(\frac{hc}{\lambda} \right) \quad (6.34)$$

The standard deviation of the intensity can also be computed from the properties of the Poisson distribution.

$$\sigma_I = \frac{1}{TA} \left(\frac{hc}{\lambda} \right) \sigma_n = \frac{\langle I \rangle}{\sqrt{TL}} = \frac{\langle I \rangle}{\sqrt{\langle n \rangle}} \quad (6.35)$$

As this equation shows, the uncertainty of getting the mean or expected intensity grows as the number of photons is reduced, a phenomenon known as *shot noise*. Perhaps a more useful form is to consider the standard deviation as a fraction of the average value.

$$\frac{\sigma_I}{\langle I \rangle} = \frac{\sigma_n}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}} = \left[\langle I \rangle TA \left(\frac{\lambda}{hc} \right) \right]^{-1/2} \quad (6.36)$$

The shot noise (the relative uncertainty in the actual intensity that the resist will see) increases with decreasing intensity, exposure time and area of concern.

As an example, consider a 193-nm exposure of a resist with a dose-to-clear of 10 mJ/cm². At the resist edge, the mean exposure energy ($=\langle I \rangle T$) will be on the order of the dose-to-clear. At this wavelength, the energy of one photon, hc/λ , is about 1.03×10^{-18} J. For an area of 1×1 nm, the mean number of photons during the exposure, from Equation (6.34), is about 97. The standard deviation is about 10, or about 10 % of the average. For an area of $10 \text{ nm} \times 10 \text{ nm}$, the number of photons increases by a factor of 100, and the relative standard deviation decreases by a factor of 10, to about 1 %. Since these are typical values for a 193-nm lithography process, we can see that shot noise contributes a noticeable amount of uncertainty as to the actual dose seen by the photoresist when looking at length scales less than about 10 nm.

For *Extreme Ultraviolet Lithography* (EUVL), the situation will be considerably worse. At a wavelength of 13.5 nm, the energy of one photon will be 1.47×10^{-17} J, about 15 times greater than at 193 nm. Also, the goal for resist sensitivity will be to have EUV resists that are 2–4 times more sensitive than 193-nm resists (though it is unclear whether this goal will be achieved). Thus, the number of photons will be 30–60 times less for EUV than for a 193-nm lithography. A 1×1 nm area will see only two to three photons, and a 100-nm^2 area will see on the order of 200 photons, with a standard deviation of 7 %.

6.4.2 Chemical Concentration

As mentioned above, there really is no such thing as concentration at a point in space since the chemical species is discrete, not continuous. Concentration, the average number of molecules per unit volume, exhibits counting statistics identical to photon emission. Let C be the average number of molecules per unit volume, and dV a volume small enough so that at most one molecule may be found in it (thus requiring that the concentration be fairly dilute, so that the position of one molecule is independent of the position of other molecules). The probability of finding a molecule in that volume is just CdV . For some larger volume V , the probability of finding exactly n molecules in that volume will be given by a binomial distribution exactly equivalent to that for photon counting. And, for any reasonably large volume ($CV > 1$), this binomial distribution will also be well approximated by a Poisson distribution.

$$P(n) = \frac{(CV)^n}{n!} e^{-CV} \quad (6.37)$$

The average number of molecules in the volume will be CV , and the variance will also be CV . The relative uncertainty in the number of molecules in a certain volume will be

$$\frac{\sigma_n}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}} = \frac{1}{\sqrt{CV}} \quad (6.38)$$

[The requirement that the concentration be ‘dilute’ can be expressed as an upper limit to the Poisson distribution – for a given molecule size, saturation occurs at some n_{max} molecules in the volume V . So long as $P(n_{\text{max}})$ is small, the mixture can be said to be dilute.⁹]

As an example, consider a 193-nm resist that has an initial PAG concentration of 3 % by weight, or a concentration of about 0.07 mole/liter (corresponding to a density of

1.2 g/ml and a PAG molecular weight of 500 g/mole). Converting from moles to molecules with Avogadro's number, this corresponds to 0.042 molecules of PAG per cubic nanometer. In a volume of $(10\text{ nm})^3$, the mean number of PAG molecules will be 42 (see Figure 6.8). The standard deviation will be 6.5 molecules, or about 15%. For 248-nm resists, the PAG loading is typically 3 times higher or more, so that closer to 150 PAG molecules might be found in a $(10\text{-nm})^3$ volume, for a standard deviation of 8%. Note that when the mean number of molecules in a given volume exceeds about 20, the Poisson distribution can be well approximated with a Gaussian distribution.

As mentioned briefly above, Poisson statistics apply only for reasonably low concentrations. The random distribution of molecules assumes that the position of each molecule is independent of all the others. As concentrations get higher, the molecules begin to 'crowd' each other, reducing their randomness. In the extreme limit, molecules become densely packed and the uncertainty in concentration goes to zero. This saturation condition is a function of not only the concentration, but the size of the molecule as well. To avoid saturation, the volume fraction occupied by the molecules under consideration must be small.

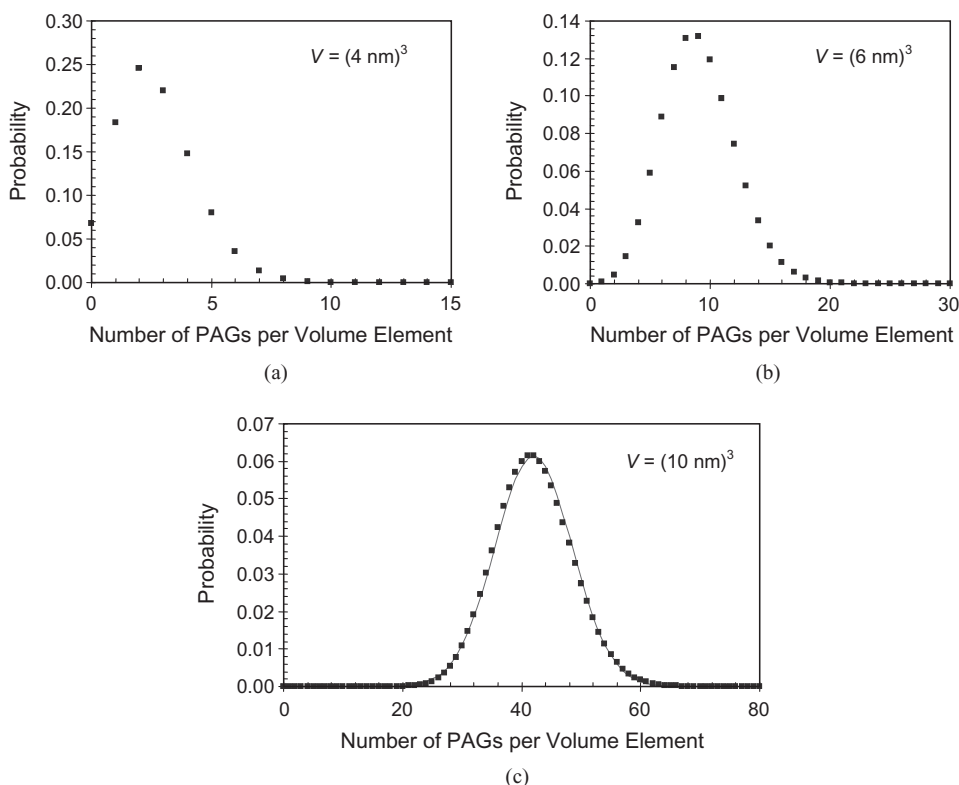


Figure 6.8 Example Poisson distributions for a typical 193-nm resist ($G_0N_A = 0.042/\text{nm}^3$): (a) volume of 64 nm^3 ; (b) volume of 216 nm^3 ; and (c) volume of 1000 nm^3 . For (c), a Gaussian distribution with the same mean and standard deviation as the Poisson distribution is shown as a solid line

6.4.3 Some Mathematics of Binary Random Variables

The mathematical approach to be taken below in considering the stochastic nature of molecular events (such as absorption, reaction or diffusion) is to define a binary random variable to represent the result of that event for a single molecule. For example, we may let y be a random variable that represents whether a given single PAG molecule was converted to acid or remains unexposed by the end of the exposure process. Thus, $y = 1$ if the event has happened, and $y = 0$ if the event has not happened. Then, by considering the kinetics of the event or some other physical model of how that event might happen, we assign a probability P to the event. Since the event is binary, $P(y = 1) = 1 - P(y = 0)$.

We next consider some properties of the random variable y . In particular, what is the mean and variance of y ? The mean value of y uses the definition of the expectation value for a discrete random variable:

$$\langle y \rangle \equiv E(y) = \sum_{i=0,1} y_i P(y_i) = (0)P(0) + (1)P(1) = P(1) \quad (6.39)$$

Likewise, the variance of y is

$$\sigma_y^2 \equiv E((y - \langle y \rangle)^2) = E(y^2) - \langle y \rangle^2 \quad (6.40)$$

But, since y is binary, $y^2 = y$ and

$$\sigma_y^2 = \langle y \rangle - \langle y \rangle^2 = \langle y \rangle(1 - \langle y \rangle) \quad (6.41)$$

Thus, if an appropriate probability for a molecular event can be assigned, the mean and variance of the binary random variable y , representing that event for a single molecule, can easily be calculated. Of course, these results are simply the properties of a binomial distribution, which is the probability distribution of a binary random variable. Interestingly, the random variable y has no uncertainty at the extremes of the event process ($\langle y \rangle = 1$ or $\langle y \rangle = 0$), and maximum uncertainty when $\langle y \rangle = 0.5$.

Now consider a collection of molecules. Suppose a volume V contains some number n molecules. After the event process is complete (e.g. at the end of exposure), the number of molecules Y in that volume that underwent the event will be

$$Y = \sum_{i=1}^n y_i \quad (6.42)$$

where y_i is the discrete random variable representing the state of the i th molecule found in this volume (each event in the volume is assumed independent). For a given n , the mean and variance of Y can be readily computed:

$$\begin{aligned} \langle Y | n \rangle &= \sum_{i=1}^n \langle y_i \rangle = n \langle y \rangle \\ \sigma_{Y|n}^2 &= \sum_{i=1}^n \sigma_{y_i}^2 = n \sigma_y^2 \end{aligned} \quad (6.43)$$

where the notation $\langle ab \rangle$ signifies the mean value of a for a given b .

But n itself is not a fixed quantity and will in general follow a Poisson statistical distribution, as discussed in section 6.4.2. To find the mean value of Y including the statistical variation of n ,

$$E(Y) = \langle Y \rangle = \sum_{n=0}^{\infty} \langle Y | n \rangle \frac{\langle n \rangle^n}{n!} e^{-\langle n \rangle} = \sum_{n=0}^{\infty} \langle y \rangle n \frac{\langle n \rangle^n}{n!} e^{-\langle n \rangle} = \langle n \rangle \langle y \rangle \quad (6.44)$$

Likewise, the variance of Y can be computed from

$$\sigma_Y^2 = E(Y - \langle Y \rangle)^2 = E(Y^2) - \langle Y \rangle^2 \quad (6.45)$$

Starting the calculation of the expectation value of Y^2 ,

$$E(Y^2) = E\left(\left[\sum_{i=1}^n y_i\right]^2\right) = E\left(\sum_{i=1}^n \sum_{j=1}^n y_i y_j\right) = E\left(\sum_{i=1}^n \sum_{j=1}^n E(y_i y_j)\right) \quad (6.46)$$

Assuming that the exposure state of each molecule is independent of the states of the others,

$$\begin{aligned} E(y_i y_j) &= E(y^2), \quad i = j \\ E(y_i y_j) &= E(y_i)E(y_j), \quad i \neq j \end{aligned} \quad (6.47)$$

giving

$$E(Y^2) = E[n(n-1)\langle y \rangle^2 + nE(y^2)] = E[n^2\langle y \rangle^2 + n\sigma_y^2] = \langle y \rangle^2 E(n^2) + \sigma_y^2 E(n) \quad (6.48)$$

But, since

$$E(n^2) = \sigma_n^2 + \langle n \rangle^2 \quad (6.49)$$

we can combine Equations (6.45), (6.48) and (6.49) to give

$$\sigma_Y^2 = \langle y \rangle^2 \sigma_n^2 + \langle n \rangle \sigma_y^2 \quad (6.50)$$

The variance of n , for a Poisson distribution, is just the mean value of n . Thus,

$$\sigma_Y^2 = \langle n \rangle (\langle y \rangle^2 + \sigma_y^2) \quad (6.51)$$

Using the variance of y from Equation (6.41),

$$\sigma_Y^2 = \langle n \rangle [\langle y \rangle^2 + \langle y \rangle (1 - \langle y \rangle)] = \langle n \rangle \langle y \rangle = \langle Y \rangle \quad (6.52)$$

Like the Poisson distribution of molecules before the event process begins, the variance of the final distribution is equal to its mean. In the sections below, these simple results will be applied to the specific chemical events of exposure, diffusion and reaction.

6.4.4 Photon Absorption and Exposure

What is the probability that a photon will be absorbed by a molecule of light-sensitive material in the resist? Further, what is the probability that a molecule of sensitizer will react to form an acid? As discussed above, there will be a statistical uncertainty in the number of photons in a given region of resist, a statistical uncertainty in the number of PAG molecules, and additionally a new statistical uncertainty in the absorption and exposure event itself.

Consider a single molecule of PAG. First-order kinetics of exposure was used to derive Equation (6.3), the concentration of PAG remaining after exposure (and, as well, the concentration of acid generated) in the continuum approximation (this is also called the *mean-field* solution to the kinetics of exposure). From a stochastic modeling perspective, this kinetic result represents a probability density function for reaction: G/G_0 is the fraction of PAG that is unreacted in some large volume, and by the Law of Large Numbers this must be the probability that any given PAG will remain unexposed. Let y be a random variable that represents whether a given single PAG molecule remains unexposed or was converted to acid by the end of the exposure process. Thus $y = 0$ means an acid has been generated (PAG has reacted), and $y = 1$ means the PAG has not been exposed (no acid generated). A kinetic analysis of exposure gives us the probability for each of these states, given a certain intensity-in-resist I :

$$P(y = 0|I) = 1 - e^{-CIt}, \quad P(y = 1|I) = e^{-CIt} \quad (6.53)$$

For a given intensity, the mean value of y can be calculated using the definition of a discrete probability expectation value:

$$\langle y|I \rangle = P(1|I) = e^{-CIt} \quad (6.54)$$

Likewise, the variance of y for a given intensity is

$$\sigma_{y|I}^2 = \langle y \rangle (1 - \langle y \rangle) = e^{-CIt} (1 - e^{-CIt}) \quad (6.55)$$

Note that the above equations were carefully derived under the assumption of a given intensity of light. However, we know from our discussion of photon counting statistics that I is a probabilistic function [the aerial image intensity $I(x)$ can be thought of as proportional to the probability of finding a photon at position x]. Thus, the mean and variance of y must take into account this probabilistic nature. Letting n be the number of photons exposing a given area A over an exposure time t [related to I by Equation (6.33)], it will be useful to define a new constant:

$$CIt = \psi n \quad \text{which leads to } \psi = \left(\frac{hc}{\lambda} \right) \frac{C}{N_A A} = \frac{\Phi a_M}{N_A A} = \frac{\Phi \sigma_{M-abs}}{A} \quad (6.56)$$

where the last two equations come from the microscopic absorption analysis presented in Chapter 5. The term ψ is the *exposure shot noise coefficient*, and is equal to the exposure quantum efficiency multiplied by the ratio of the PAG absorption cross section to the area of statistical interest. Since the exposure quantum efficiency is typically in the 0.3–0.7 range and the PAG absorption cross section is on the order of $1 \text{ \AA}^2$ for 193-nm resists, for most areas of interest this exposure shot noise coefficient will be much less than 1.

Using this exposure shot noise coefficient to convert intensity to number of photons, the total probability becomes

$$P(y \text{ and } n) = P(y|n)P(n) = \begin{cases} (1 - e^{-\psi n}) \frac{\langle n \rangle^n}{n!} e^{-\langle n \rangle} & \text{for } y = 0 \\ e^{-\psi n} \frac{\langle n \rangle^n}{n!} e^{-\langle n \rangle} & \text{for } y = 1 \end{cases} \quad (6.57)$$

$$\langle y \rangle = \sum_{n=0}^{\infty} \sum_y y P(y \text{ and } n) = \sum_{n=0}^{\infty} (e^{-\psi})^n \frac{\langle n \rangle^n}{n!} e^{-\langle n \rangle} \quad (6.58)$$

Evaluating the summation (and recognizing the Taylor series expansion of an exponential),

$$\langle y \rangle = \sum_{n=0}^{\infty} \frac{(e^{-\psi} \langle n \rangle)^n}{n!} e^{-\langle n \rangle} = e^{-\langle n \rangle (1 - e^{-\psi})} \quad (6.59)$$

When ψ is small, the exact expression (6.59) can be approximated as

$$\begin{aligned} \langle y \rangle &= e^{-\langle n \rangle (1 - e^{-\psi})} \approx e^{-\langle n \rangle (\psi - 0.5\psi^2)} \\ &= [e^{-\langle n \rangle \psi}]^{1 - \psi/2} = [e^{-C(I)t}]^{1 - \psi/2} = e^{-C(I)t} e^{\psi^2 \langle n \rangle / 2} = e^{-C(I)t} e^{\frac{1}{2} (Ct\sigma_I)^2} \end{aligned} \quad (6.60)$$

This final approximate form can be derived in an alternate way. For a reasonable number of photons (greater than about 20), the Poisson distribution of photons is well approximated by a normal distribution of the same mean and variance. A normal distribution for I will result in a log-normal distribution for y . The final equation on the right-hand side of Equation (6.60) is the well-known result for the mean of a log-normal distribution.

As often happens when taking statistical distributions into account, the mean value of the output of the function is not equal to the function evaluated at the mean value of the input. Another approximate form for the mean value of y is

$$\langle y \rangle \approx y_c + \frac{\psi}{2} (y_c) \ln(y_c) \quad \text{where } y_c = e^{-C(I)t} \quad (6.61)$$

Since $(y_c) \ln(y_c)$ is always negative (it goes between zero and about -0.37), the mean value of y is always less than the value of the function y evaluated at the mean value of the intensity. When averaging over a large area, ψ goes to zero, so that y_c can be thought of as the mean value of y when averaged over a large area (that is, in the continuum limit). As the area over which we average our results decreases, the mean value of y (which is the same as the mean value of the relative acid concentration, as we will see below) decreases due to shot noise.

The mean value and uncertainty of the state of *one* acid molecule after exposure can now be translated into a mean and uncertainty of the overall acid concentration after exposure. Consider a volume V that initially contains some number $n_{0-\text{PAG}}$ PAG molecules. After exposure, the number of photogenerated acid molecules Y will be

$$Y = \sum_{i=1}^{n_{0-\text{PAG}}} y_i \quad (6.62)$$

Assuming that each exposure event is independent, the results from section 6.4.3 give the mean of Y :

$$\langle Y \rangle = \langle n_{0-\text{PAG}} \rangle \langle y \rangle \quad (6.63)$$

The variance becomes

$$\sigma_Y^2 = \langle Y \rangle + \langle Y \rangle^2 (e^{(Ct\sigma_I)^2} - 1) = \langle Y \rangle + \langle Y \rangle^2 (e^{\psi^2 \langle n \rangle} - 1) \quad (6.64)$$

Since ψ will in general be small,

$$\sigma_Y^2 \approx \langle Y \rangle + \langle Y \rangle^2 (\psi^2 \langle n \rangle) = \langle Y \rangle + \langle Y \rangle^2 (C \langle I \rangle t) \psi = \langle Y \rangle + \frac{\langle Y \rangle^2 (C \langle I \rangle t)^2}{\langle n \rangle} \quad (6.65)$$

The variance of Y has two components. The Poisson chemical distribution gives the first term, $\langle Y \rangle$. Photon shot noise adds a second term, inversely proportional to the mean number of photons.

At this point it is useful to relate the number of acid molecules per unit volume Y to the concentration of acid H , and the initial number of PAGs $n_{0\text{-PAG}}$ to the initial PAG concentration G_0 .

$$H = G_0 - \frac{Y}{N_A V}, \quad G_0 = \frac{n_{0\text{-PAG}}}{N_A V} \quad (6.66)$$

We can also define a relative acid concentration h to be

$$h = \frac{H}{\langle G_0 \rangle} = \frac{G_0}{\langle G_0 \rangle} - \frac{Y}{\langle n_{0\text{-PAG}} \rangle} = \frac{n_{0\text{-PAG}}}{\langle n_{0\text{-PAG}} \rangle} - \frac{Y}{\langle n_{0\text{-PAG}} \rangle} \quad (6.67)$$

The means of these quantities can be related by

$$\langle h \rangle = \frac{\langle H \rangle}{\langle G_0 \rangle} = 1 - \frac{\langle Y \rangle}{\langle n_{0\text{-PAG}} \rangle} = 1 - \langle y \rangle \quad (6.68)$$

Using Equation (6.67),

$$\sigma_h^2 = \frac{\langle h \rangle}{\langle n_{0\text{-PAG}} \rangle} + \frac{\sigma_Y^2 - \langle Y \rangle}{\langle n_{0\text{-PAG}} \rangle^2} \quad (6.69)$$

Finally, using Equation (6.65), the variance in acid concentration will be

$$\sigma_h^2 = \frac{\langle h \rangle}{\langle n_{0\text{-PAG}} \rangle} + \frac{[(1 - \langle h \rangle) \ln(1 - \langle h \rangle)]^2}{\langle n \rangle} \quad (6.70)$$

This final result, which accounts for photon fluctuations, uncertainty in the initial concentration of photoacid generator, and the probabilistic variations in the exposure reaction itself, is reasonably intuitive. The first term on the right-hand side of Equation (6.70) is the expected Poisson result based on exposure kinetics – the relative uncertainty in the resulting acid concentration after exposure goes as one over the square root of the mean number of acid molecules generated within the volume of interest. For large volumes and reasonably large exposure doses, the number of acid molecules generated is large and the statistical uncertainty in the acid concentration becomes small. For small volumes or low doses, a small number of photogenerated acid molecules results in a large uncertainty in the actual number within that volume. The second term accounts for photon shot noise. For the case of the $(10 \text{ nm})^3$ of 193-nm resist given above, the standard deviation in initial acid concentration near the resist edge (where the mean acid concentration will be about 0.4) will be >20%. For 193-nm resists, the impact of photon shot noise is minimal compared to variance in acid concentration caused by simple molecular position uncertainty.

6.4.5 Acid Diffusion, Conventional Resist

In Chapter 5, diffusion was treated using a Gaussian diffusion point spread function (DPSF) – the latent image after diffusion is the initial latent image convolved with a Gaussian, whose standard deviation σ is called the diffusion length. From a stochastic modeling perspective, the Gaussian represents a probability distribution: for a particle initially at some position, the Gaussian *DPSF*, given by

$$DPSF = (2\pi\sigma^2)^{-3/2} e^{-r^2/2\sigma^2} \quad (6.71)$$

represents the probability density of finding that particle some distance r from its original location. This probability distribution is itself derived from a stochastic look at the possible motions of the particle during the bake. Given that the particle can randomly move in any possible direction at a particular speed determined by its diffusivity, and can change directions randomly, the resulting path of the particle is called a *random walk*. Averaging over all possible random walk paths produces the Gaussian probability distribution (when the diffusivity is constant).

How does diffusion affect the statistical fluctuations of concentration? The latent image of acid after exposure, $h(x,y,z)$, as used in the continuum approximation, is actually the mean relative concentration $\langle h \rangle$, with a variance given by Equation (6.70). Ignoring the photon shot noise, the probability distribution for the number of acids in a given volume is approximately Poisson. Now consider a PEB process that causes only diffusion of acid (such as for a conventional resist). The mean acid concentration will ‘diffuse’ as was described in the continuum approximation of Chapter 5, convolving the initial acid latent image with the *DPSF*. The stochastic uncertainty in acid concentration will still be given by Equation (6.70), but with a new mean ‘after diffusion’ concentration of acid $\langle h^* \rangle$. Ignoring photon shot noise,

$$\sigma_{h^*} = \sqrt{\frac{\langle h^* \rangle}{N_A V}}, \quad \langle h^* \rangle = \langle h \rangle \otimes DPSF \quad (6.72)$$

The proof of this result (which will be sketched briefly here) becomes evident by considering first the diffusion of a single acid molecule, located at some initial position designated by the subscript i . Defining a binary random variable y_i to represent whether that molecule is found in some small volume dV located a distance r_i from its original location,

$$P(y_i = 1) = (2\pi\sigma^2)^{-3/2} e^{-r_i^2/2\sigma^2} dV \quad (6.73)$$

This binary random variable will follow all of the properties derived in section 6.4.3:

$$\langle y_i \rangle = P(y_i = 1), \quad \sigma_{y_i}^2 = \langle y_i \rangle (1 - \langle y_i \rangle) \quad (6.74)$$

If, instead of one acid molecule at a certain location which then diffuses, there are n_i acid molecules at this location that then diffuse, the total number of acid molecules in that volume dV will be Y_i , again with the properties derived in section 6.4.3:

$$Y_i = \sum_{k=1}^{n_i} y_{ik}, \quad \langle Y_i \rangle = \langle n_i \rangle \langle y_i \rangle, \quad \sigma_{Y_i}^2 = \langle Y_i \rangle \quad (6.75)$$

Adding up the contributions from all of the locations that could possibly contribute acid molecules into the volume dV produces the convolution result of Equation (6.72).

$$Y = \sum_i Y_i$$

$$\langle Y \rangle = \sum_i \langle Y_i \rangle = \sum_i \langle n_i \rangle \langle y_i \rangle = \sum_i \langle n_i \rangle (2\pi\sigma^2)^{-3/2} e^{-r_i^2/2\sigma^2} dV \quad (6.76)$$

Here, a summation over i is equivalent to an integration over all space, so that the final summation becomes an integral, which in this case becomes a convolution.

$$\langle Y \rangle = \langle n \rangle \otimes DPSF \quad (6.77)$$

How does diffusion affect the variance of the post-diffusion concentration?

$$\sigma_Y^2 = E(Y^2) - \langle Y \rangle^2$$

$$E(Y^2) = E\left(\left[\sum_i Y_i\right]^2\right) = E\left(\sum_i Y_i \sum_j Y_j\right) = \sum_i \sum_j E(Y_i Y_j) = \sum_i \sum_j E\left(\sum_{k=1}^{n_i} \sum_{l=1}^{n_j} E(y_{ik} y_{jl})\right) \quad (6.78)$$

Since the diffusion of a given molecule is independent of all of the other molecules,

$$E(y_{ik} y_{jl}) = E(y_{ik}^2), \quad ik = jl$$

$$E(y_{ik} y_{jl}) = E(y_{ik})E(y_{jl}), \quad ik \neq jl \quad (6.79)$$

As a result, the variance follows in the same way as shown in section 6.4.3:

$$\sigma_Y^2 = \sum_i \sigma_{Y_i}^2 = \langle Y \rangle \quad (6.80)$$

Thus, diffusion does not intrinsically increase the uncertainty in acid concentration due to the extra stochastic process of the random walk, nor does it intrinsically ‘smooth out’ any uncertainties through the process of diffusion. Only through a change in the mean concentration does diffusion affect the uncertainty in the concentration. For a typical conventional resist, the concentration of exposure products near the edge of the resist feature does not change appreciably due to diffusion during PEB. Thus, it is unlikely that diffusion will have a significant impact on the concentration statistics near the resist line edge.

6.4.6 Acid-Catalyzed Reaction–Diffusion

Of course, for a chemically amplified resist acid diffusion is accompanied by one or more reactions. In this section and the next, we’ll consider only the polymer deblocking reaction (acid-quencher neutralization will be discussed in a following section). In the continuum limit, the amount of blocked polymer left after the PEB is given by

$$M = M_0 e^{-K_{\text{amp}} t_{\text{PEB}} h_{\text{eff}}}$$

$$h_{\text{eff}}(x, y, z) = \frac{1}{t_{\text{PEB}}} \int_0^{t_{\text{PEB}}} (h(x, y, z, t=0) \otimes DPSF) dt = h(x, y, z, t=0) \otimes RDPSF \quad (6.81)$$

As before, the latent image of acid after exposure, $h(x, y, z, t=0)$ used in the continuum approximation is actually the mean acid concentration $\langle h \rangle$, with a standard deviation given

above. The effective acid concentration, however, has a very specific interpretation: it is the time average of the acid concentration at a given point. The interesting question to be answered, then, is whether this time-averaging effect of diffusion coupled with the acid-catalyzed reaction affects the uncertainty in the effective acid concentration compared to the original acid concentration uncertainty.

To determine the statistical properties of the effective acid concentration, we'll begin as we did for the pure diffusion case in the previous section by looking at the diffusion of a single molecule of acid. Let the binary random variable $y_i(t)$ represent whether that molecule is found in some small volume dV located a distance r_i from its original location, during the interval of time between t and $t + dt$.

$$P(y_i(t)=1) = (2\pi\sigma^2)^{-3/2} e^{-r_i^2/2\sigma^2} dV, \quad \sigma^2 = 2Dt \quad (6.82)$$

For n_i acid molecules at this location that then diffuse, the total number of acid molecules in that volume dV and over the same time interval will be $Y_i(t)$, again with the properties derived in the previous section:

$$Y_i(t) = \sum_{j=1}^{n_i} y_{ij}(t), \quad \langle Y_i(t) \rangle = \langle n_i \rangle \langle y_i(t) \rangle, \quad \sigma_{Y_i}^2 = \langle Y_i(t) \rangle \quad (6.83)$$

Adding up the contributions from all of the locations that could possibly contribute acid molecules into the volume dV during the interval of time between t and $t + dt$ produces the standard convolution result:

$$Y(t) = \sum_i Y_i(t) \quad (6.84)$$

$$\langle Y(t) \rangle = \sum_i \langle n_i \rangle \langle y_i(t) \rangle = \langle n \rangle \otimes DPSF(t)$$

So far, the results are identical to that for pure diffusion since we have only considered how many acids are in the volume at time t . We now wish to integrate over time, from 0 to t_{PEB} .

$$Y = \frac{1}{t_{PEB}} \int_0^{t_{PEB}} Y(t) dt \quad (6.85)$$

$$\langle Y \rangle = \frac{1}{t_{PEB}} \int_0^{t_{PEB}} \langle Y(t) \rangle dt = \frac{1}{t_{PEB}} \int_0^{t_{PEB}} \langle n \rangle \otimes DPSF(t) dt = \langle n \rangle \otimes RDPSF$$

Thus, as expected, the effective acid concentration used in the continuum approximation is in fact the mean value of a stochastic random variable. The uncertainty of Y , however, involves some extra complications. Proceeding as we have done in the past,

$$\sigma_Y^2 = E(Y^2) - \langle Y \rangle^2$$

$$E(Y^2) = \frac{1}{t_{PEB}^2} E \left(\left[\int_0^{t_{PEB}} Y(t) dt \right]^2 \right) = \frac{1}{t_{PEB}^2} E \left(\int_0^{t_{PEB}} Y(t) dt \int_0^{t_{PEB}} Y(t') dt' \right) \quad (6.86)$$

$$= \frac{1}{t_{PEB}^2} \int_0^{t_{PEB}} \int_0^{t_{PEB}} E(Y(t')Y(t)) dt dt'$$

What is different in this case compared to previous derivations is that $Y(t')$ and $Y(t)$ are not independent. If an acid finds itself in the volume of interest at time t , the probability of finding that same acid in the same volume at time t' will be greater than if the acid had not been found in the volume at time t . Thus, there will be some correlation between $Y(t')$ and $Y(t)$ that will be strongest when t' is close to t , and diminish as the difference between the two times increases.

Applying the definition of $Y(t)$ to Equation (6.86),

$$E(Y^2) = \frac{1}{t_{PEB}^2} \int_0^{t_{PEB}} \int_0^{t_{PEB}} E \left(\sum_i^{n_i} y_{ij}(t) \sum_k^{n_k} y_{kl}(t') \right) dt dt' \quad (6.87)$$

For $ij \neq kl$ (that is, for different acid molecules), there can be no correlation through time. Thus, the only time correlation comes from the same acid molecule venturing into the volume at different times. Changing the order of integration to bring the time integration inside and the spatial integration to the outside, we are interested in finding

$$\Lambda = \frac{1}{t_{PEB}^2} \int_0^{t_{PEB}} \int_0^{t_{PEB}} E \left(\sum_{j=1}^{n_i} y_{ij}(t) \sum_{l=1}^{n_i} y_{il}(t') \right) dt dt' \quad (6.88)$$

Since the only correlation through time exists when $j = l$,

$$E \left(\sum_{j=1}^n y_j(t) \sum_{l=1}^n y_l(t') \right) = E[n(n-1)\langle y(t) \rangle \langle y(t') \rangle + nE(y(t)y(t'))] \quad (6.89)$$

Following along the same lines as the derivation given in section 6.4.3, we obtain

$$\begin{aligned} \sigma_Y^2 &= \langle n \rangle \otimes CovPSF \\ CovPSF &= \frac{1}{t_{PEB}^2} \int_0^{t_{PEB}} \int_0^{t_{PEB}} E(y(t)y(t')) dt dt' \end{aligned} \quad (6.90)$$

where *CovPSF* is a new function that I call the ‘covariance point spread function’.

Consider the unphysical case of $y(t)$ and $y(t')$ being perfectly correlated. In that case, $y(t)y(t') = y(t)$ (since y is a binary variable) and the *CovPSF* is equal to the *RDPSF*. Thus, perfect correlation over the bake time produces the result that

$$\sigma_Y^2 = \langle n \rangle \otimes RDPSF = \langle Y \rangle \quad (6.91)$$

indicating that the final effective acid concentration would retain an approximately Poisson distribution. Without perfect correlation, the *CovPSF* will differ from the *RDPSF* and the variance will be different from the Poisson result.

Since $y(t)$ is a binary random variable, the expectation value is equal to the probability that both conditions are true (that is, the acid found its way into the volume at both time t and time t').

$$E(y(t')y(t)) = P(y(t') = 1 \text{ and } y(t) = 1) = P(y(t') = 1 | y(t) = 1)P(y(t) = 1) \quad (6.92)$$

The conditional probability is governed by the difference in time between t and t' .

$$P(y(t') = 1 | y(t) = 1) = \frac{e^{-r^2/2\sigma^2}}{\sqrt{2\pi\sigma}} dV, \quad \sigma^2 = 2D|t - t'| \quad (6.93)$$

where the one-dimensional case is shown for convenience.

Thus, the *CovPSF* can be written as

$$CovPSF = \frac{dV}{4\pi Dt_{PEB}^2} \int_0^{t_{PEB}} \frac{e^{-r^2/4Dt}}{\sqrt{t}} dt \int_0^{t_{PEB}} \frac{e^{-r^2/4D|t-t'|}}{\sqrt{|t-t'|}} dt' \quad (6.94)$$

where again the 1D version is shown for convenience. The inner integral over t' can be readily evaluated:

$$\frac{1}{\sqrt{4\pi Dt_{PEB}}} \int_0^{t_{PEB}} \frac{e^{-r^2/4D|t-t'|}}{\sqrt{|t-t'|}} dt' = \frac{t_{PEB}-t}{t_{PEB}} RDPSF(t_{PEB}-t) + \frac{t}{t_{PEB}} RDPSF(t) \quad (6.95)$$

Unfortunately, the next integral does not appear to have an analytical solution. Some properties of the integral, however, can be determined. When $r = 0$, the integral evaluates to

$$CovPSF(r=0) = \frac{dV}{\sigma^2} \left(\frac{1}{2} + \frac{1}{\pi} \right), \quad \sigma^2 = 2Dt_{PEB} \quad (6.96)$$

A numerical integration shows that the 1D *CovPSF* is very similar to the *RDPSF*, but with the diffusion length replaced by $\sigma/2$:

$$CovPSF(r) \approx \frac{dV}{2\sigma} \left[4 \frac{e^{-2x^2/\sigma^2}}{\sqrt{2\pi}\sigma} - \frac{4|x|}{\sigma^2} \operatorname{erfc} \left(\frac{\sqrt{2}|x|}{\sigma} \right) \right] = \frac{dV}{2\sigma} RDPSF \left(\frac{\sigma}{2} \right) \quad (6.97)$$

A comparison of this approximate form with a numerical integration of Equation (6.94) is shown in Figure 6.9.

The impact of the *CovPSF* can now be determined. The shape of the *CovPSF* is very similar to that of the *RDPSF*, but with an effective diffusion length that is less by a factor of 2. Since the effective acid concentration near the line edge does not differ appreciably

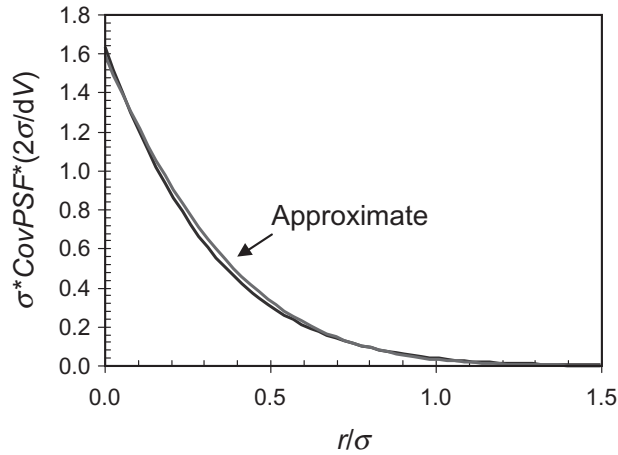


Figure 6.9 Comparison of the approximate 1D *CovPSF* of Equation (6.97) with a numerical evaluation of the defining integral in Equation (6.94)

from the acid concentration when convolved with the *RDPSF*, the same will be true when convolved with this portion of the *CovPSF*. Thus, the effective acid concentration and its standard deviation can be approximated as

$$\begin{aligned}\langle h_{\text{eff}} \rangle &= \langle h \rangle \otimes \text{RDPSF} \\ \sigma_{h_{\text{eff}}} &\approx \sqrt{\frac{dV}{2\sigma_D}} \sigma_h\end{aligned}\quad (6.98)$$

The rest of the *CovPSF* simply gives the ratio $dV/2\sigma_D$, where σ_D is the overall diffusion length. In this context, dV is the (one-dimensional) capture volume for the deblocking reaction. In other words, $dV/2$ represents how close the acid must come to a blocked site before it can potentially participate in a deblocking reaction (called the capture range, a). As Equation (6.98) indicates, when the acid diffusion length exceeds the capture range of the deblocking reaction, the impact of the diffusing catalyst is to ‘smooth out’ the statistical fluctuations present in the original acid distribution.

While the above derivation used the one-dimensional form of the diffusion probability, the extension to three dimensions is straightforward. For a problem of dimensionality p ($p = 1, 2$ or 3),

$$\sigma_{h_{\text{eff}}} \approx \left(\frac{a}{\sigma_D} \right)^{\frac{p}{2}} \sigma_h \quad (6.99)$$

where a is the deblocking reaction capture distance. In other words, if the acid diffuses a distance less than the reaction capture range, the catalytic nature of the amplification reaction actually increases the stochastic variation in the effective acid concentration compared to the original acid concentration. If, however, the diffusion length is greater than this capture range, the time-averaging effect of the catalytic reaction will smooth out stochastic roughness. It is not diffusion, *per se*, that reduces stochastic uncertainty, but rather the diffusion of a reaction catalyst that does so. Since in real resist systems the diffusion length will invariably be greater than the reaction capture distance, the net affect will always be a reduction in the effective acid concentration standard deviation.

6.4.7 Reaction–Diffusion and Polymer Deblocking

The stochastics of the deblocking of a single blocked site will follow along the same lines as the single PAG exposure analysis of section 6.4.4. Let y be a random variable that represents whether a given single blocked site remains blocked by the end of the PEB. Thus $y = 1$ means the site remains blocked, and $y = 0$ means the site has been deblocked. As before, the continuum kinetic analysis gives us the probability that a single site is deblocked for a given effective acid concentration.

$$P(y = 1 | h_{\text{eff}}) = e^{-K_{\text{amp}} t_{\text{PEB}} h_{\text{eff}}}, \quad P(y = 0 | h_{\text{eff}}) = 1 - e^{-K_{\text{amp}} t_{\text{PEB}} h_{\text{eff}}} \quad (6.100)$$

The probability distribution of h_{eff} , however, is not obvious. While the relative acid concentration has a Poisson distribution, the time-averaging effect on the acid diffusion turns the discrete acid random variable into a continuous effective acid random variable.

It will be reasonable to assume that h_{eff} is normally distributed with mean and standard deviations as given in the previous section. Thus, the mean value of y becomes

$$\begin{aligned}\langle y \rangle &= \frac{1}{\sqrt{2\pi}\sigma_{h_{\text{eff}}}} \int_{-\infty}^{\infty} (e^{-K_{\text{amp}} t_{\text{PEB}} h_{\text{eff}}}) e^{-(h_{\text{eff}} - \langle h_{\text{eff}} \rangle)^2 / 2\sigma_{h_{\text{eff}}}^2} dh_{\text{eff}} \\ \langle y \rangle &= e^{-K_{\text{amp}} t_{\text{PEB}} \langle h_{\text{eff}} \rangle} e^{\frac{1}{2}(K_{\text{amp}} t_{\text{PEB}} \sigma_{h_{\text{eff}}})^2}\end{aligned}\quad (6.101)$$

The random variable y has a log-normal probability distribution and Equation (6.101) can be recognized as the standard result for a log-normal distribution.

The total number of blocked groups remaining in a certain small volume will be given by Y .

$$Y = \sum_{i=1}^{n_{0-\text{block}}} y_i \quad (6.102)$$

The mean of Y can be easily computed, as before.

$$\langle Y \rangle = \langle n_{0-\text{block}} \rangle \langle y \rangle \quad (6.103)$$

The variance of Y can be found with a result similar to that for photon shot noise during exposure:

$$\sigma_Y^2 = \langle Y \rangle + \langle Y \rangle^2 (e^{(K_{\text{amp}} t_{\text{PEB}} \sigma_{h_{\text{eff}}})^2} - 1) \quad (6.104)$$

From the definitions of M and Y ,

$$\langle M_0 \rangle = \frac{\langle n_{0-\text{blocked}} \rangle}{N_A V}, \quad \langle M \rangle = \frac{\langle Y \rangle}{N_A V}, \quad m \equiv \frac{M}{\langle M_0 \rangle} \quad (6.105)$$

Thus,

$$\langle m \rangle = \langle y \rangle \quad (6.106)$$

and

$$\sigma_m^2 = \frac{\sigma_M^2}{\langle M_0 \rangle^2} = \frac{\sigma_Y^2}{\langle n_{0-\text{blocked}} \rangle^2} \quad (6.107)$$

giving

$$\sigma_m^2 = \frac{\langle m \rangle}{\langle n_{0-\text{blocked}} \rangle} + \langle m \rangle^2 (e^{(K_{\text{amp}} t_{\text{PEB}} \sigma_{h_{\text{eff}}})^2} - 1) \quad (6.108)$$

For small levels of effective acid uncertainty,

$$\sigma_m^2 \approx \frac{\langle m \rangle}{\langle n_{0-\text{blocked}} \rangle} + \langle m \rangle^2 (K_{\text{amp}} t_{\text{PEB}} \sigma_{h_{\text{eff}}})^2 = \frac{\langle m \rangle}{\langle n_{0-\text{blocked}} \rangle} + (\langle m \rangle \ln \langle m \rangle)^2 \left(\frac{\sigma_{h_{\text{eff}}}}{\langle h_{\text{eff}} \rangle} \right)^2 \quad (6.109)$$

As before, the first term captures the Poisson uncertainty due to the initial distribution of blocked polymer. The second term captures the influence of the effective acid concentration uncertainty. Combining this expression with Equation (6.99) for the variance of the effective acid concentration,

$$\sigma_m^2 = \frac{\langle m \rangle}{\langle n_{0\text{-blocked}} \rangle} + (\langle m \rangle \ln \langle m \rangle)^2 \left(\frac{\sigma_h}{\langle h \rangle} \right)^2 \left(\frac{a}{\sigma_D} \right)^p \quad (6.110)$$

Finally, using Equation (6.70) for the variance of the acid concentration,

$$\sigma_m^2 = \frac{\langle m \rangle}{\langle n_{0\text{-blocked}} \rangle} + \left(\frac{\langle m \rangle \ln \langle m \rangle}{\langle h \rangle} \right)^2 \left(\frac{a}{\sigma_D} \right)^p \left(\frac{\langle h \rangle}{\langle n_{0\text{-PAG}} \rangle} + \frac{[(1 - \langle h \rangle) \ln(1 - \langle h \rangle)]^2}{\langle n \rangle} \right) \quad (6.111)$$

Or, in a slightly different form,

$$\left(\frac{\sigma_m}{\langle m \rangle} \right)^2 = \frac{1}{\langle n_{0\text{-blocked}} \rangle \langle m \rangle} + (K_{\text{amp}} t_{\text{PEB}})^2 \left(\frac{a}{\sigma_D} \right)^p \left(\frac{\langle h \rangle}{\langle n_{0\text{-PAG}} \rangle} + \frac{[(1 - \langle h \rangle) \ln(1 - \langle h \rangle)]^2}{\langle n \rangle} \right) \quad (6.112)$$

While the above equations show how fundamental parameters affect the resulting variance in the final blocked polymer concentration, interpretation is somewhat complicated by the fact that these parameters are not always independent. In particular, the Byers–Petersen model of Equation (6.26) shows a relationship between $K_{\text{amp}} t_{\text{PEB}}$ and $\sigma_D a$. For the case where the PEB temperature puts the deblocking reaction in the diffusion-controlled regime, Equation (6.28) can be expressed as

$$K_{\text{amp}} t_{\text{PEB}} = 2\pi \sigma_D^2 a \langle G_0 \rangle N_A \quad (6.113)$$

When used in Equation (6.112), for the 3D case ($p = 3$), and ignoring photon shot noise, the result is

$$\begin{aligned} \left(\frac{\sigma_m}{\langle m \rangle} \right)^2 &= \frac{1}{\langle n_{0\text{-blocked}} \rangle \langle m \rangle} + (2\pi \sigma_D^2 a \langle G_0 \rangle N_A)^2 \left(\frac{a}{\sigma_D} \right)^3 \left(\frac{\langle h \rangle}{\langle n_{0\text{-PAG}} \rangle} \right) \\ &= \frac{1}{\langle n_{0\text{-blocked}} \rangle \langle m \rangle} + (2\pi)^2 \left(\frac{\sigma_D a^5}{V^2} \right) \langle h \rangle \langle n_{0\text{-PAG}} \rangle \\ &= \frac{1}{V} \left[\frac{1}{\langle M_0 \rangle N_A \langle m \rangle} + (2\pi)^2 \sigma_D a^5 \langle h \rangle \langle G_0 \rangle N_A \right] \end{aligned} \quad (6.114)$$

While this result is interesting, it is unlikely that resist will be processed at temperatures high enough to put the PEB reaction fully into the diffusion-controlled regime.

Using the example of a typical 193-nm resist, $M_0 N_A = 1.2/\text{nm}^3$, $G_0 N_A = 0.042/\text{nm}^3$ and $K_{\text{amp}} t_{\text{PEB}} = 2$. Consider the case of $\langle h \rangle = \langle h_{\text{eff}} \rangle = 0.3$, $\sigma_D/9 = 5$ and $p = 3$. For a $(10 \text{ nm})^3$ volume, $\sigma_h/\langle h \rangle \approx 0.28$ and $\sigma_{h_{\text{eff}}}/\langle h_{\text{eff}} \rangle \approx 0.025$. The remaining blocked polymer will have $\langle m \rangle = 0.55$ and $\sigma_m = 0.023$, or about 4.3 %. For a $(5 \text{ nm})^3$ volume, $\sigma_m = 0.064$, or about 11 %. The factors that have the largest impact are the diffusion length and the volume of resist under examination, as shown in Figure 6.10.

6.4.8 Acid–Base Quenching

Interestingly, the acid–base neutralization reaction poses the greatest challenge to stochastic modeling. While acid concentrations in chemically amplified resists are low, base quencher concentrations are even lower, leading to greater statistical uncertainty in

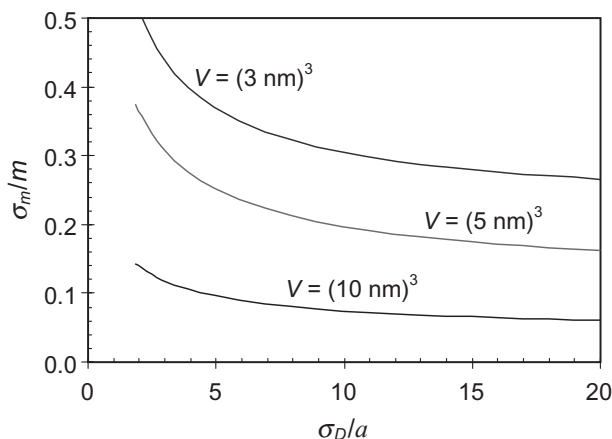


Figure 6.10 The relative uncertainty of the relative blocked polymer concentration as a function of the volume (= length cubed) under consideration and the diffusion length of acid for a two-dimensional problem. Typical 193-nm resist values were used: $G_0N_A = 0.042/\text{nm}^3$, $M_0N_A = 1.2/\text{nm}^3$, $K_{amp}t_{PEB} = 2$, and for an exposure such that $\langle h \rangle = 0.4$

concentration for small volumes. Further, since the reaction is one of annihilation, statistical variations in acid and base concentrations can lead effectively to acid–base segregation, with clumps of all acid or all base.¹⁰ Such clumping is likely to lead to low-frequency line edge roughness that is easily discernable from the higher-frequency roughness that will be predicted with the stochastic models presented above. Further discussion of this interesting and difficult topic is beyond the scope of this book. Thus, while acid–base quenching is extremely important in its impact on LER, it will not be considered in the model being developed here.

The stochastic descriptions of exposure and reaction–diffusion in chemically amplified resist will be used in Chapter 9 to develop a model for line-edge roughness.

Problems

- 6.1. Equation (6.9) provides the (no diffusion or acid loss) kinetic solution for the blocking group concentration in a chemically amplified resist. Under the assumption that h is locally constant,
 - (a) Derive an expression for the relative bake time sensitivity of m (i.e. calculate $dm/d\ln t_{PEB}$).
 - (b) Derive an expression for the relative temperature sensitivity of m (i.e. calculate $dm/d\ln T$). From this, will a low activation energy resist or a high activation energy resist be more sensitive to temperature variations?
 - (c) Does the presence of base quencher change the bake time or temperature sensitivity of m ?
- 6.2. Plot the effective thermal dose versus temperature for a resist with an amplification reaction $E_a = 25.0 \text{ kcal/mol}$ and $A_r = \exp(28.0) \text{ s}^{-1}$, and an exposure rate constant $C = 0.05 \text{ cm}^2/\text{mJ}$. Use a temperature range from 100 to 140 °C and assume a PEB time of 60 seconds.

- 6.3. What is the impact of increasing the PAG loading (the initial concentration of PAG) on the effective thermal dose for a given PEB process?
- 6.4. Derive the simplified Fujita–Doolittle Equation (6.15) from the full Fujita–Doolittle equation given in Chapter 5. What assumptions have to be made?
- 6.5. Ignoring diffusion and acid loss mechanisms, derive an expression for the catalytic chain length (the number of deblocking events per acid molecule). At what dose will this catalytic chain length be at its maximum? What is the maximum value for the catalytic chain length?
- 6.6. One possible acid loss mechanism is called acid trapping, where acid is consumed by being trapped within the resist. Ignoring diffusion and all other acid loss mechanisms, this acid loss mechanism is governed by

$$\frac{\partial H}{\partial t_{\text{PEB}}} = -k_s H$$

Under these conditions, derive an expression for the concentration of blocked sites at the end of the PEB.

- 6.7. For $x \gg \sigma_D$, derive a simplified ‘large argument’ approximation to the 1D reaction–diffusion point spread function.
- 6.8. Henry’s law (first formulated in 1803) states that the amount of a gas which will dissolve into a solution is directly proportional to the partial pressure of that gas above the solution, and is generally found to be true only at low concentrations. When applied to a solid, the amount of a substance that is adsorbed onto the surface of the solid (the surface concentration) is proportional to the partial pressure of the substance in the atmosphere. The constant of proportionality, called the Henry’s law constant, is dependent on both materials. Suppose ammonia is in the atmosphere at a concentration of 100 ppb (parts per billion), and that the Henry’s law constant for ammonia in photoresist is 30 liters-atm/mol at room temperature. What is B_0 , the surface concentration of ammonia?
- 6.9. Derive an expression for the total amount of base contaminant found inside a resist, assuming that Equation (6.22) applies and that $\sigma_B \ll$ resist thickness. If one wanted to reduce this total amount of base contaminant by a factor of 2, how much would the post-exposure delay time have to be reduced (all other factors held constant)?
- 6.10. Why does the addition of base quencher reduce the sensitivity of the resist to airborne base contaminants?
- 6.11. For the deblocking kinetics data of Figure 6.7, assume a resist with an exposure rate constant $C = 0.03 \text{ cm}^2/\text{mJ}$, and a PEB time of 60 seconds.
 - (a) If the resist shows no deblocking until a dose of $8 \text{ mJ}/\text{cm}^2$, what is the relative quencher loading for the resist?
 - (b) Estimate the amplification rate constant for each temperature shown.
 - (c) What is the activation energy for the amplification reaction?
- 6.12. Assuming that a certain amount of acid is required to achieve a desired lithographic effect (that is, assuming the mean concentration of photogenerated acid is fixed), how low can the mean number of photons go before photon shot noise exceeds the PAG loading shot noise? Assume, for example, that $\langle h \rangle = 0.3$ and $\langle n_{0-\text{PAG}} \rangle / V = 0.05/\text{nm}^3$, and the region of interest is $(10 \text{ nm})^3$.

- 6.13. Suppose for a given process that $\langle h \rangle = 0.3$ and $\langle m \rangle = 0.5$. Further, assume that photon shot noise can be neglected and that $\langle n_{0\text{-blocked}} \rangle / \langle n_{0\text{-PAG}} \rangle = 30$. Considering the total uncertainty of the final blocked polymer concentration for a 1D problem, how large must σ_D/a be in order to make the contribution of acid uncertainty equal to the Poisson uncertainty of the deblocking reaction?
- 6.14. As Equation (6.111) shows, increasing acid diffusion during PEB reduces the stochastic uncertainty caused by acid concentration fluctuations. Is there a limit as to how much this uncertainty can be reduced (that is, is there a practical maximum value for the acid diffusion length)? Explain.
- 6.15. Qualitatively, how will the presence of base quencher affect the variance of final blocked polymer concentration?

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